

Aprendizaje de Máquina

U2 Análisis de Clústeres

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Section 1

Introduction

Hierarchical clustering

Hierarchical clustering algorithms are of different philosophy from BSAS. Specifically, instead of producing a single clustering, they produce a **hierarchy of clusterings**.

This kind of algorithm is usually found in the social sciences and biological taxonomy.

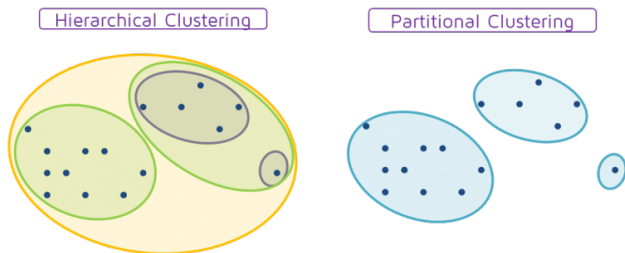


Figure: Hierarchical clustering vs partitional clustering

Nested clustering

Let $X = \{x_i, i = 1, \dots, N\}$ be a set of l -dimensional vectors that are to be clustered.

We represent a clustering as

$$R = \{C_j, j = 1, \dots, m\}$$

where $C_j \subseteq X$.

A clustering R_1 containing k clusters is said to be **nested** in the clustering R_2 , which contains $r < k$, if **each** cluster in R_1 is a subset of some set (cluster) in R_2 . In this case we write

$$R_1 \sqsubset R_2$$

Nested clustering

Note that at least one cluster of R_1 is a proper subset of R_2 .

For example, the clustering

$$R_1 = \{\{x_1, x_3\}, \{x_4\}, \{x_2, x_5\}\}$$

is nested in

$$R_2 = \{\{x_1, x_3, x_4\}, \{x_2, x_5\}\}$$

On the other hand, R_1 is nested neither in

$$R_3 = \{\{x_1, x_4\}, \{x_3\}, \{x_2, x_5\}\}$$

$$R_4 = \{\{x_1, x_2, x_4\}, \{x_3, x_5\}\}$$

It is clear that a clustering is not nested to itself.

Hierarchical clustering

Hierarchical clustering algorithms produce a **hierarchy of nested clusterings**.

More specifically, these algorithms involve N steps, as many as the number of data vectors.

At each step t , a new clustering is obtained based on the clustering produced at the previous step $t - 1$.

There are two main categories of these algorithms:

- **Agglomerative**
- **Divisive**

Hierarchical clustering

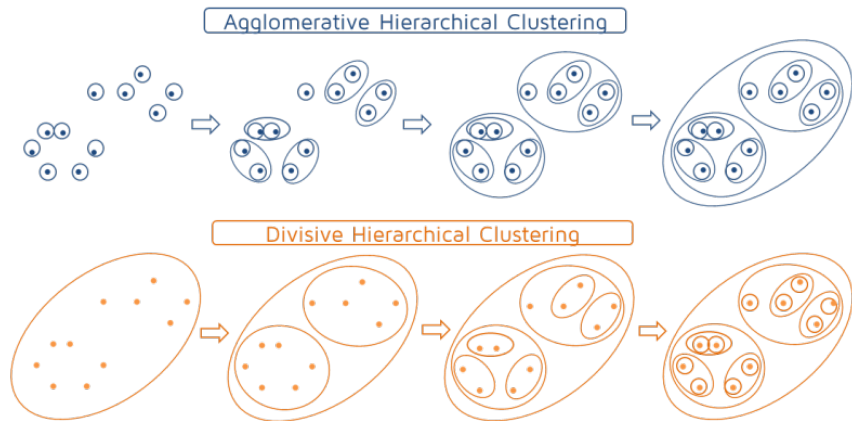


Figure: Outline of agglomerative and divisive algorithms

Agglomerative algorithms

- The initial clustering R_0 for the **agglomerative algorithms** consists of N clusters, each containing a single element of X .
- At the first step, the clustering R_1 is produced. It contains $N - 1$ sets, such that

$$R_0 \sqsubset R_1$$

- This procedure continues until the final clustering, R_{N-1} , is obtained, which contains a single set, that is, the set of data, X .
- Notice that for the hierarchy of the resulting clusterings, we have

$$R_0 \sqsubset R_1 \sqsubset \dots \sqsubset R_{N-1}$$

Divisive algorithms

- The **divisive algorithms** follow the inverse path.
- In this case, the initial clustering R_0 consists of a single set, X .
- At the first step the clustering R_1 is produced. It consists of two sets, such that $R_1 \sqsubset R_0$.
- This procedure continues until the final clustering R_{N-1} is obtained, which contains N sets, each consisting of a single element of X .
- In this case we have

$$R_{N-1} \sqsubset R_{N-2} \sqsubset \dots \sqsubset R_0$$

Section 2

Agglomerative algorithms

Let $g(C_i, C_j)$ be a function defined for **all possible pairs** of clusters of X .

This function measures the **proximity** between C_i and C_j .

Let t denote the current **level of hierarchy**.

Generalized Agglomerative Scheme (GAS)

- Initialization:

- Choose $R_0 = \{C_i = \{x_i\}, i = 1, \dots, N\}$ as the initial clustering
- $t = 0$

- Repeat:

- $t = t + 1$
- Among all possible pairs of clusters (C_r, C_s) in R_{t-1} find the one, say (C_i, C_j) such that
 - If g is a **dissimilarity** function

$$g(C_i, C_j) = \min_{r,s} g(C_r, C_s)$$

- If g is a **similarity** function

$$g(C_i, C_j) = \max_{r,s} g(C_r, C_s)$$

- Define $C_q = C_i \cup C_j$ and produce the new clustering

$$R_t = (R_{t-1} - \{C_i, C_j\}) \cup \{C_q\}$$

- Until all vectors lie in a single cluster.

Agglomerative Scheme

It is clear that this scheme creates a hierarchy of N clusterings, so that each one is nested in all successive clusterings, that is,

$$R_{t_1} \sqsubset R_{t_2}, \quad \text{for } t_1 < t_2, \quad t_2 = 1, \dots, N-1$$

Alternatively, we can say that if two vectors come together into a single cluster at level t of the hierarchy, they will **remain in the same cluster** for all subsequent clusterings. This is another way of viewing the **nesting property**.

Agglomerative Scheme

A disadvantage of the **nesting property** is that there is no way to recover from a *poor* clustering that may have occurred in an earlier level of the hierarchy.

The **total number of operations** required by an agglomerative scheme is proportional to N^3 . However, the exact complexity of the algorithm depends on the definition of g .

Pattern matrix

The **pattern matrix** $D(X)$ is the $N \times I$ matrix, whose i th row is the (transposed) i th vector of X .

The **similarity (dissimilarity) matrix**, $P(X)$, is an $N \times N$ matrix whose (i, j) element equals the similarity $s(x_i, x_j)$ (dissimilarity $d(x_i, x_j)$) between vectors x_i and x_j . It is also referred to as the **proximity matrix** to include both cases.

In general, P is a **symmetric matrix**. Moreover,

- If P is a similarity matrix, its diagonal elements are equal to the maximum value of s .
- On the other hand, if P is a dissimilarity matrix, its diagonal elements are equal to the minimum value of d .

Example

Let $X = \{x_i, i = 1, \dots, 5\}$, with $x_1 = (1, 1)^T$, $x_2 = (2, 1)^T$, $x_3 = (5, 4)^T$, $x_4 = (6, 5)^T$.

The pattern matrix of X is

$$D(X) = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 5 & 4 \\ 6 & 5 \end{bmatrix}$$

What is the dissimilarity matrix when the L_1 and L_∞ are in use?

Dendrogram

A **threshold dendrogram**, or simply a **dendrogram**, is an effective means of representing the **sequence of clustering**.

A tree-like diagram called a dendrogram is often used to visualize the hierarchy of clusters. It displays the order in which clusters have been merged or divided and shows the similarity or distance between data points.

Each step of the **generalized agglomerative scheme (GAS)** corresponds to a level of the dendrogram.

Cutting the dendrogram at a specific level results in a clustering.

Dendrogram

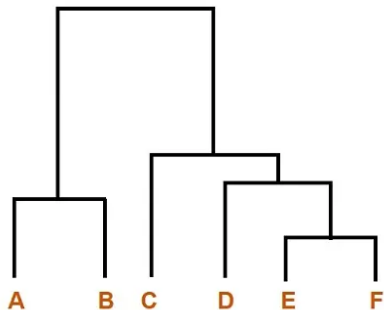
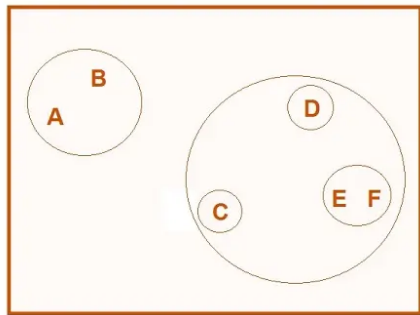


Figure: Dendrogram example

Dendrogram

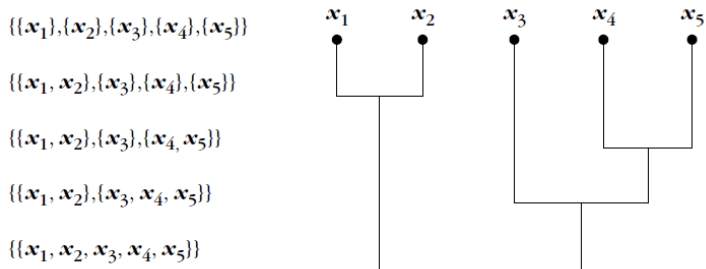


Figure: Dendrogram example

Agglomerative Algorithms based on Matrix Theory

These algorithms may be viewed as **special cases** of GAS.

The **input** in these schemes is the $N \times N$ dissimilarity matrix, $P_0 = P(X)$, derived from X .

At each level, t , when two clusters are merged into one, the size of the dissimilarity matrix $_t$ becomes $(N - t) \times (N - t)$.

P_t follows from P_{t-1} by

- 1 **Deleting** the two rows and columns that correspond to the merged clusters
- 2 **Adding** a new row and a new column that contains the distances between the newly formed cluster and the old (unaffected at this level) clusters.

Matrix Updating Algorithm Scheme (MUAS)

- Initialization:

- $R_0 = \{\{\mathbf{x}_i\}, i = 1, \dots, N\}$
- $P_0 = P(X)$
- $t = 0$

- Repeat:

- $t = t + 1$
- Find C_i, C_j such that

$$d(C_i, C_j) = \min_{r,s=1,\dots,N, r \neq s} d(C_r, C_s)$$

- Merge C_i, C_j into a single cluster C_q and form
 $R_t = (R_{t-1} - \{C_i, C_j\}) \cup \{C_q\}$
- Define the proximity matrix P_t from P_{t-1}
- Until R_{N-1} clustering is formed, that is, all vectors lie in the same cluster.

Update Algorithms

- The **single link algorithm**:

$$d(C_q, C_s) = \min\{d(C_i, C_s), d(C_j, C_s)\}$$

The single link has a tendency to favor **elongated clusters**.

- The **complete link algorithm**:

$$d(C_q, C_s) = \max\{d(C_i, C_s), d(C_j, C_s)\}$$

The complete link algorithm proceeds by recovering small **compact clusters**.

- The **Ward** or **minimum variance** algorithm. Here, the distance between two clusters C_i and C_j , d_{ij} , is defined as a weighted version of the squared Euclidean distance of their mean vectors, that is,

$$d_{ij} = \frac{n_i n_j}{n_i + n_j} \|m_i - m_j\|^2$$

where $d_{ij} = \|m_i - m_j\|^2$.

Single and complete link algorithm

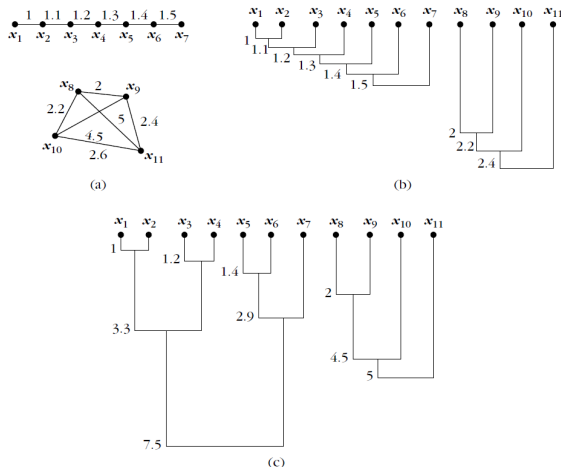


Figure: (a) The data set X . (b) The dissimilarity dendrogram produced by the single link algorithm. (c) The dissimilarity dendrogram produced by the complete link algorithm.

References

- Theodoridis, S. & Koutroumbas, K. *Pattern Recognition*, 4th ed., Academic Press, 2009